

ZEPTO

Quick Commerce Data Analysis

SQL-Driven Product & Inventory Intelligence Report

3,000+

SKUs Analysed

12+

SQL Queries

5

Analysis Domains

Prepared by: Data Analyst | Tools: MySQL
Domain: E-Commerce Analytics

1. Project Overview

This report presents a comprehensive SQL-based analysis of Zepto's product catalogue and inventory data. Zepto is one of India's fastest-growing quick-commerce platforms, offering 10-minute grocery and household deliveries. The analysis covers data exploration, cleaning, pricing analytics, discount intelligence, and inventory optimisation across all product categories.

Attribute	Details
Platform	Zepto — Quick Commerce (10-minute delivery)
Dataset	Product catalogue with SKU, pricing, discount, weight, stock data
Database	MySQL (Serial Primary Key, Numeric, Boolean types)
Key Table	zepto (sku_id, category, name, mrp, discountPercent, availableQuantity, discountedSellingPrice, weightInGms, outOfStock, quantity)
Skills Used	DDL, DML, Aggregation, CASE logic, Data Cleaning

2. Database Design & Schema

2.1 Table Structure

The zepto table was designed with appropriate data types to handle product catalogue data efficiently. Key design decisions include using SERIAL for auto-incrementing primary key, NUMERIC for financial precision, and BOOLEAN for stock status.

```
CREATE TABLE zepto (  
  sku_id          SERIAL PRIMARY KEY,  
  category        VARCHAR(150),  
  name            VARCHAR(150) NOT NULL,  
  mrp              NUMERIC(8,2),  
  discountPercent NUMERIC(5,2),  
  availableQuantity INT,  
  discountedSellingPrice NUMERIC(8,2),  
  weightInGms      INT,  
  outOfStock       BOOLEAN,  
  quantity         INT  
);
```

2.2 Schema Insights

- SERIAL PRIMARY KEY ensures each SKU has a unique, auto-generated identifier
- NUMERIC(8,2) for MRP and price fields prevents floating-point errors in financial calculations
- BOOLEAN for outOfStock enables fast filtering and aggregation on stock status
- VARCHAR(150) for category and name accommodates Zepto's diverse product naming conventions

3. Data Exploration

3.1 Initial Exploration Queries

The first phase involved understanding the dataset structure, row counts, category diversity, and identifying any data quality issues before analysis.

```
-- Row count & category overview
SELECT COUNT(*) FROM zepto;           -- Total SKUs
SELECT COUNT(DISTINCT category) FROM zepto;  -- Unique categories

-- Check for NULL values across all columns
SELECT * FROM zepto
WHERE name IS NULL OR category IS NULL OR mrp IS NULL
    OR discountPercent IS NULL OR availableQuantity IS NULL
    OR discountedSellingPrice IS NULL OR weightInGms IS NULL
    OR outOfStock IS NULL OR quantity IS NULL;

-- Product duplication check
SELECT name, COUNT(sku_id) FROM zepto
GROUP BY name HAVING COUNT(sku_id) > 1
ORDER BY COUNT(sku_id) DESC;
```

3.2 Exploration Findings

Metric	Finding	Implication
Total SKUs	3,000+ records	Large enough for statistical relevance
Unique Categories	Multiple distinct categories	Rich multi-category analysis possible
NULL Values	None found	Dataset is clean and complete
Duplicate Products	Some names appear multiple times	Different SKUs for size variants
Price Format	Values in paise (pre-cleaning)	Required conversion to INR

3.3 Stock Status Distribution



In-Stock vs Out-of-Stock

Analysis revealed a significant portion of products are out of stock. This directly impacts revenue potential and customer experience, suggesting supply chain gaps in certain categories.

```
-- Stock status breakdown
SELECT outOfStock, COUNT(sku_id) AS product_count
FROM zepto
GROUP BY outOfStock;
```

4. Data Cleaning

4.1 Removing Zero-Price Products

Products with MRP = 0 are invalid entries — either data entry errors or system placeholders. These were identified and removed to ensure pricing analysis integrity.

```
-- Identify zero-price products
SELECT * FROM zepto WHERE mrp = 0;

-- Remove invalid entries
DELETE FROM zepto WHERE mrp = 0;
```

4.2 Currency Conversion: Paise to Rupees

A critical data quality issue was discovered: all price values were stored in paise (1 INR = 100 paise) instead of rupees. A bulk UPDATE was executed to normalise all financial values.

```
-- Convert paise to INR (₹) for all price columns
UPDATE zepto
SET mrp = mrp / 100.0,
    discountedSellingPrice = discountedSellingPrice / 100.0;
```

Cleaning Step	Method	Impact
Zero-price records	DELETE WHERE mrp = 0	Eliminated invalid SKUs
Currency unit fix	UPDATE SET mrp = mrp/100	Corrected all financial metrics
NULL check	WHERE col IS NULL	Confirmed data completeness
Table rename	ALTER TABLE ... RENAME TO	Standardised table naming

5. Pricing & Discount Analysis

5.1 Top 10 Best Value Products

Products offering the highest discount percentage represent the best deals for customers and the highest clearance rate for Zepto. These are typically loss leaders or promotional items.

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
ORDER BY discountPercent DESC
LIMIT 10;
```



Insight: High-Discount Strategy

Products with discount percentages above 40% are likely promotional anchors or overstocked items. Zepto uses aggressive discounting in categories like snacks, beverages, and personal care to drive basket size and customer retention.

5.2 Premium Products with Low Discounts

Identifying high-MRP products (>₹500) with less than 10% discount helps spot categories where Zepto maintains full margin — typically premium or specialty products.

```
SELECT name, mrp, discountPercent
FROM zepto
WHERE mrp > 500 AND discountPercent < 10
ORDER BY mrp DESC;
```

5.3 Top 5 Categories by Average Discount

```
SELECT category,
       ROUND(AVG(discountPercent), 2) AS avg_discount
FROM zepto
GROUP BY category
ORDER BY avg_discount DESC
LIMIT 5;
```

#	Typical Category Type	Business Rationale
1	Snacks & Munchies	High volume, competitive market
2	Cold Drinks & Juices	Drive trial & repeat purchase
3	Breakfast & Dairy	Daily essentials — price sensitive
4	Personal Care	Brand competition forces discounts
5	Instant / Ready Foods	Clearance of near-expiry stock

6. Inventory & Revenue Analysis

6.1 Estimated Revenue by Category

Revenue potential is calculated as $\text{discountedSellingPrice} \times \text{availableQuantity}$. This gives a conservative estimate of maximum achievable revenue from current in-stock inventory.

```
SELECT category,
       SUM(discountedSellingPrice * availableQuantity) AS estimated_revenue
FROM zepto
GROUP BY category
ORDER BY estimated_revenue DESC;
```



Revenue Concentration Risk

If the top 3 categories contribute more than 60% of estimated revenue, Zepto faces concentration risk. Diversifying high-margin categories and ensuring consistent stock in revenue-driving segments is critical for financial stability.

6.2 High-Value Out-of-Stock Products

Products with MRP > ₹300 that are out of stock represent direct revenue leakage. Each out-of-stock premium product is a missed conversion opportunity.

```
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE AND mrp > 300
ORDER BY mrp DESC;
```

6.3 Total Inventory Weight by Category

```
SELECT category,
       SUM(weightInGms * availableQuantity) AS total_weight_gms
FROM zepto
GROUP BY category
ORDER BY total_weight_gms DESC;
```



Warehouse Load Insight

Total inventory weight reveals which categories dominate warehouse space. Categories with high weight-to-revenue ratios are logistically expensive. Optimising SKU mix toward lighter, higher-value items improves margin per sq. ft. of dark store space.

7. Value & Weight Analytics

7.1 Price per Gram Analysis

Calculating price-per-gram enables customers and buyers to make objective comparisons across different package sizes. Products with the lowest price/gram offer the best unit value.

```
SELECT DISTINCT name,
    discountedSellingPrice,
    weightInGms,
    ROUND(discountedSellingPrice / weightInGms, 2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram DESC;
```

7.2 Product Weight Segmentation

Products are classified into Low (<1kg), Medium (1-5kg), and Bulk (>5kg) segments. This segmentation enables targeted promotions — bulk buyers respond to volume discounts while low-weight buyers prioritise convenience.

```
SELECT DISTINCT name, weightInGms,
CASE
    WHEN weightInGms <= 1000 THEN 'Low (<1kg)'
    WHEN weightInGms <= 5000 THEN 'Medium (1-5kg)'
    ELSE 'Bulk (>5kg)'
END AS weight_category
FROM zepto;
```

Segment	Weight Range	Typical Products
Low	Up to 1,000g	Snacks, spices, sachets
Medium	1,001g – 5,000g	Rice, flour, cooking oil
Bulk	Above 5,000g	Large pack staples, water, detergent

8. Key Business Insights & Recommendations

1 Fix Out-of-Stock Premium Products

High-MRP products ($>₹300$) that are out-of-stock represent the highest revenue leakage per unit. Prioritise restocking these SKUs first — even 100 units of a ₹500 product = ₹50,000 unlocked revenue.

2 Audit Extreme Discounts

Products with discounts above 50% may indicate pricing errors, expiry-driven clearance, or margin erosion. A quarterly discount audit using the top-discount query can flag anomalies before they impact P&L.

3 Optimise Category Mix for Revenue

Revenue by category ($\text{discountedSellingPrice} \times \text{qty}$) should guide procurement decisions. Categories with high revenue potential but high out-of-stock rates deserve dedicated inventory buffers.

4 Use Weight Segmentation for Logistics

Bulk items ($>5\text{kg}$) consume disproportionate dark-store shelf space. If their revenue contribution is low, consider reducing SKU count in this segment and redirecting space to high-velocity medium-weight products.

5 Price/Gram Benchmark for Procurement

Use price-per-gram as a procurement benchmark. When multiple suppliers offer the same product, the price/gram metric enables objective comparison and negotiation leverage.

9. Complete SQL Query Reference

9.1 Query Summary Table

#	Query Purpose	Key Clauses	Output Type
1	Row count	COUNT(*)	Scalar
2	Distinct categories	COUNT(DISTINCT)	Scalar
3	NULL value check	IS NULL, OR	Row set
4	Duplicate products	GROUP BY, HAVING	Ranked list
5	Zero-price removal	DELETE WHERE	DML action
6	Currency normalisation	UPDATE SET	DML action
7	Top 10 discount products	ORDER BY, LIMIT	Top-N list
8	High MRP, low discount	WHERE AND	Filtered set
9	Category revenue estimate	SUM, GROUP BY	Aggregated
10	Out-of-stock premium items	WHERE, ORDER BY	Priority list
11	Price per gram	ROUND, division	Ranked metric
12	Weight segmentation	CASE WHEN	Segmented set
13	Inventory weight by category	SUM, GROUP BY	Aggregated

10. Conclusion

This project demonstrates the end-to-end capability of SQL for real-world business intelligence. Starting from raw transactional data, the analysis progressed through schema design, data quality assurance, and business-focused insight generation — mirroring the workflow of a professional data analyst in an e-commerce environment.

The Zepto dataset is ideal for SQL portfolio projects because it combines multiple analytical dimensions: pricing, discounting, inventory, and logistics. Each query in this project solves a concrete business problem, making the analysis directly applicable to real e-commerce roles.

Data Quality NULL checks, price fixes, deduplication	Business Insights Revenue, discounts, stock gaps	SQL Mastery DDL, DML, aggregation, CASE, joins
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